SINGAPORE MANAGEMENT UNIVERSITY



OPIM 201 Operations Management

MATERIAL REQUIREMENT PLANNING (MRP)

& ENTERPRISE RESOURCE PLANNING (ERP)

Outline

- MRP Introduction
- MRP Setup
- ERP Benefits and Pitfalls
- ERP Latest Trends

Forecasting → Inventory Policy → Production Plan

- Apple forecasts the demand for ipad mini. (Forecasting)
- Based on the forecast, Apple decides on how many Ipad mini to produce/order. (EOQ or Newsvendor)
- Based on order information from Apple, Foxconn decides on the production plan of components needed. (MRP & ERP)

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FREE GROUND SHIPPING ON ALL BIKES, SCOOTERS, RIDE-ONS PURCHASED FROM HUFFYBIKES.COM! (AVAILABLE ONLY FOR THE CONTINENTAL 48 STATES)

Check out these 2019 summer favorites below.



Deluxe™ Women's Cruiser Bike, Blue, 26-inch

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Carom™ Men's Gravel Bike, Black, 27.5-inch

SHOP NOW



Sienna™ Women's Comfort Bike, White, 27.5-inch

SHOP NOW



Casoria™ Men's Comfort Bike, Blue, 27.5-inch

SHOP NOW

Motivating Example

- Suppose you are an assembly plant manager for Huffy mountain bicycles. Weekly demand for men and women mountain bicycles are given as follows:

Week	1	2	3	4	5	6	7	8	Total
Men	20	40	30	0	80	20	0	60	250
Women	60	25	40	50	80	20	90	35	400

On-hand inventory: 200 units of men mountain bicycles; 100 units of women mountain bicycles; 100 bicycle wheels;

- How many bicycle frames do you need for Week 5?
- How many bicycle wheels do you need for Week 5?
- Can you produce as much as you need?
- When do you start production?

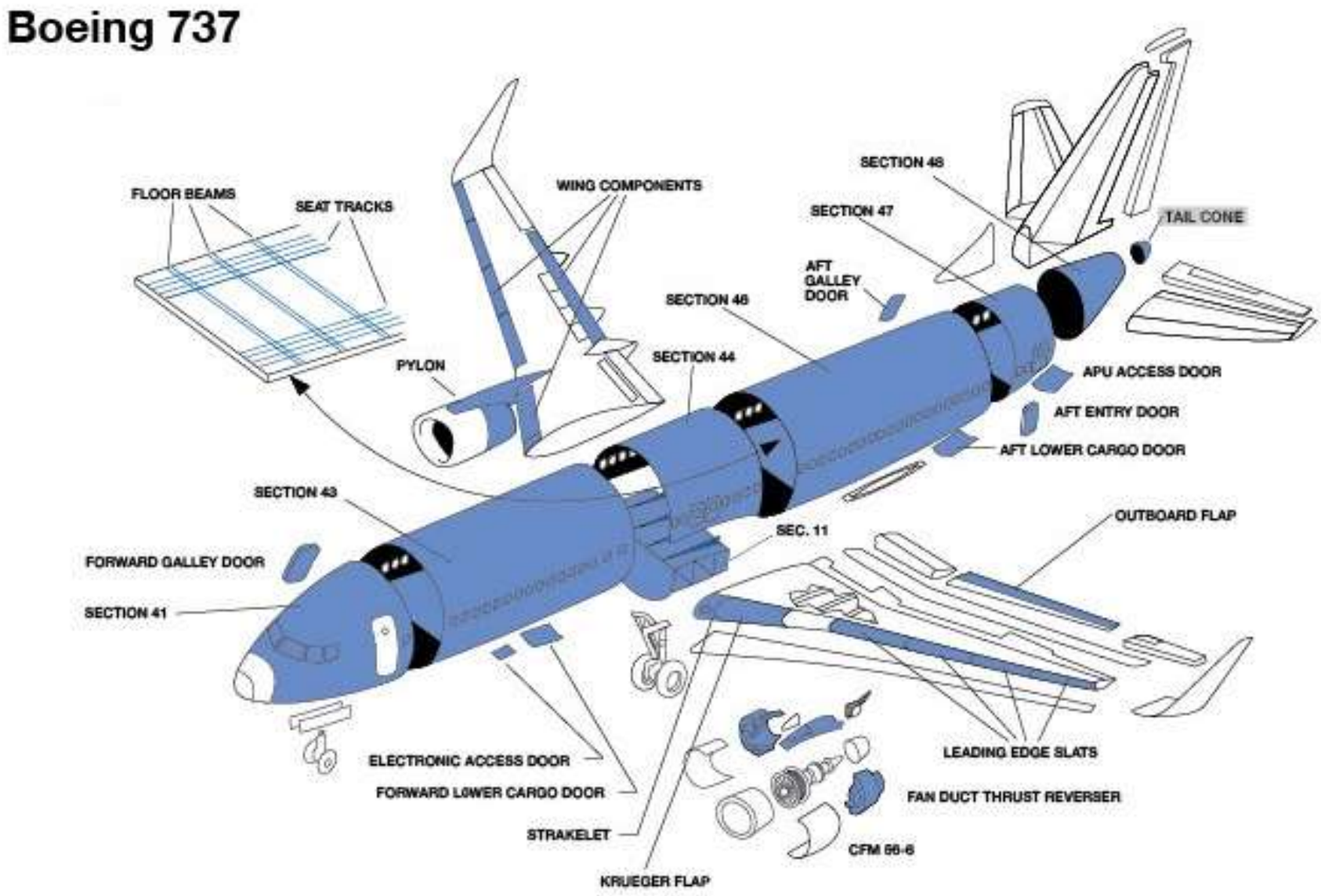


men



women

Boeing 737 Has Almost 400,000 Parts

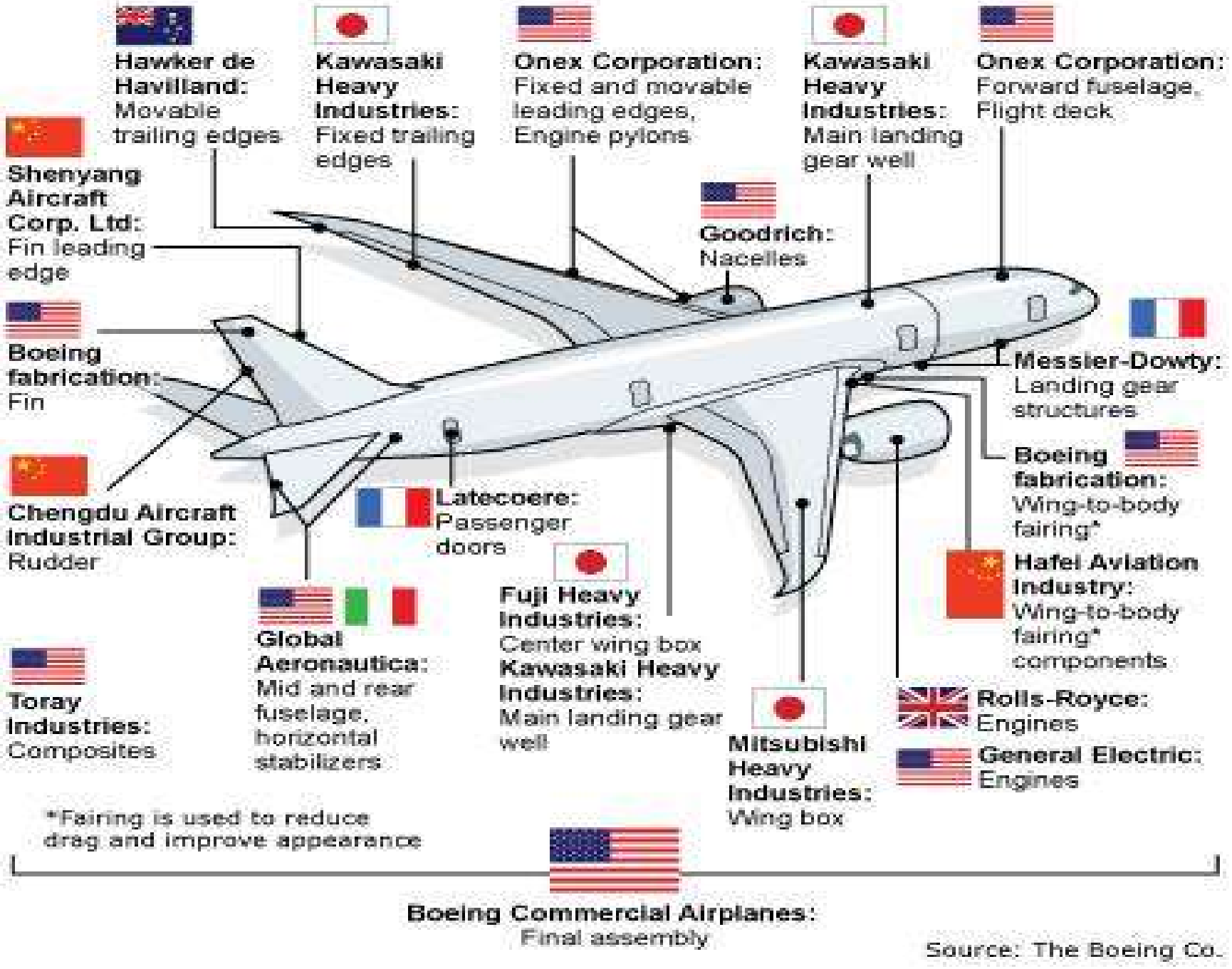


Some of the International Suppliers of Boeing 737 Parts

- Aircraft manufacturing is complex & involves production & outsourcing of many parts.

SUPPLIER	HEADQUARTERS COUNTRY	COMPONENT
Latecoere	France	Passenger doors
Labinel	France	Wiring
Messier-Bugatti	France	Electric brakes
Diehl	Germany	Interior lighting
Cobham	UK	Fuel pumps and valves
Rolls-Royce	UK	Engines
Alenia Aeronautica	Italy	Upper center fuselage and horizontal stabilizers
Toray Industries	Japan	Carbon fiber for wing and tail units
Mitsubishi Heavy Industries	Japan	Wing box
Hafei Aviation	China	Parts
Korean Airlines	South Korea	Wingtips
Saab	Sweden	Cargo and access doors

787 structure suppliers



Source: The Boeing Co.

MRP in Service Industry

- Imagine that you are the manager of Starwood Hotels and Resorts Worldwide.
  - How would you determine the amount of soap, towel, food, and beverage?
  - How many staff (receptionist, concierge, house-keeping)?
  - How many equipment (fax, TV, gym exercise machines)?



MRP in Service Industry (cont'd)

- Restaurants
  - Use MRP to manage the preparation of a dish/meal
  - e.g. Veal Picante; a top selling entree in a New Orleans restaurant (see next slide)
- Houston's Park Plaza hospital & its suppliers
  - use MRP for scheduling & management of expensive surgical inventory (equipment, materials & supplies)
- Marriott Hotel
  - use MRP for renovation of hotel rooms (to compute requirement for materials, furniture & decorations)
- SMU (ERP)

Material Requirement Planning (MRP)

- Computer-based information system that translates Master Production Schedule (MPS - derived from aggregate planning) requirements for end items into time-phased requirements for subassemblies, components, and raw materials. It has evolved into what is now more commonly known as Enterprise Resource Planning (ERP).
- Place orders and plan production & inventories for "components" (whose demand is certain once MPS of end items are confirmed), taking into account
  - Forecasts for end-item demand
  - Procurement/Production lead times
  - Demand and supply variability
  - Batch sizing/lot sizing (due to setup costs)
- Good Demand Forecasts for "components"
  - What?
  - How much?
  - When?

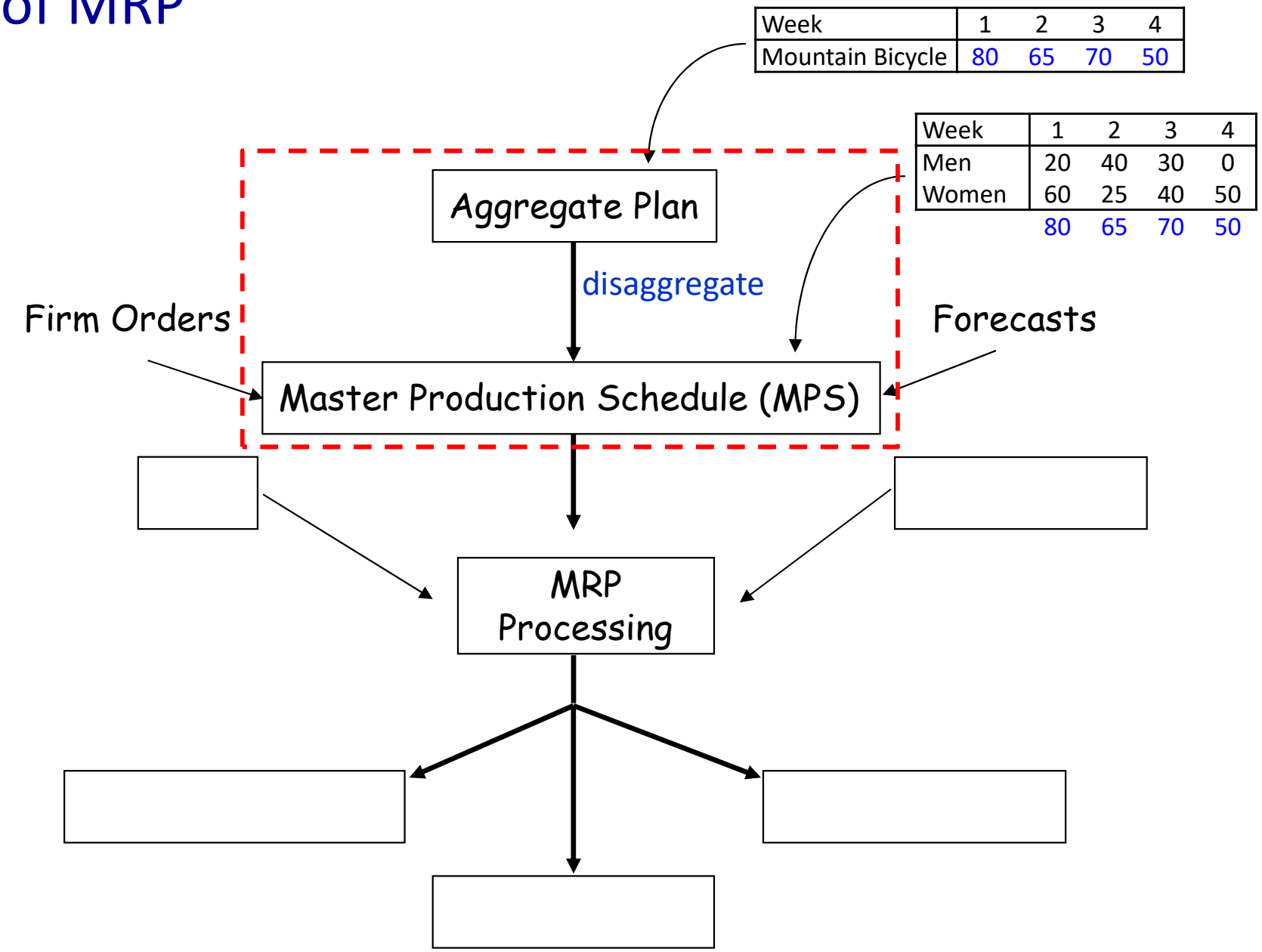






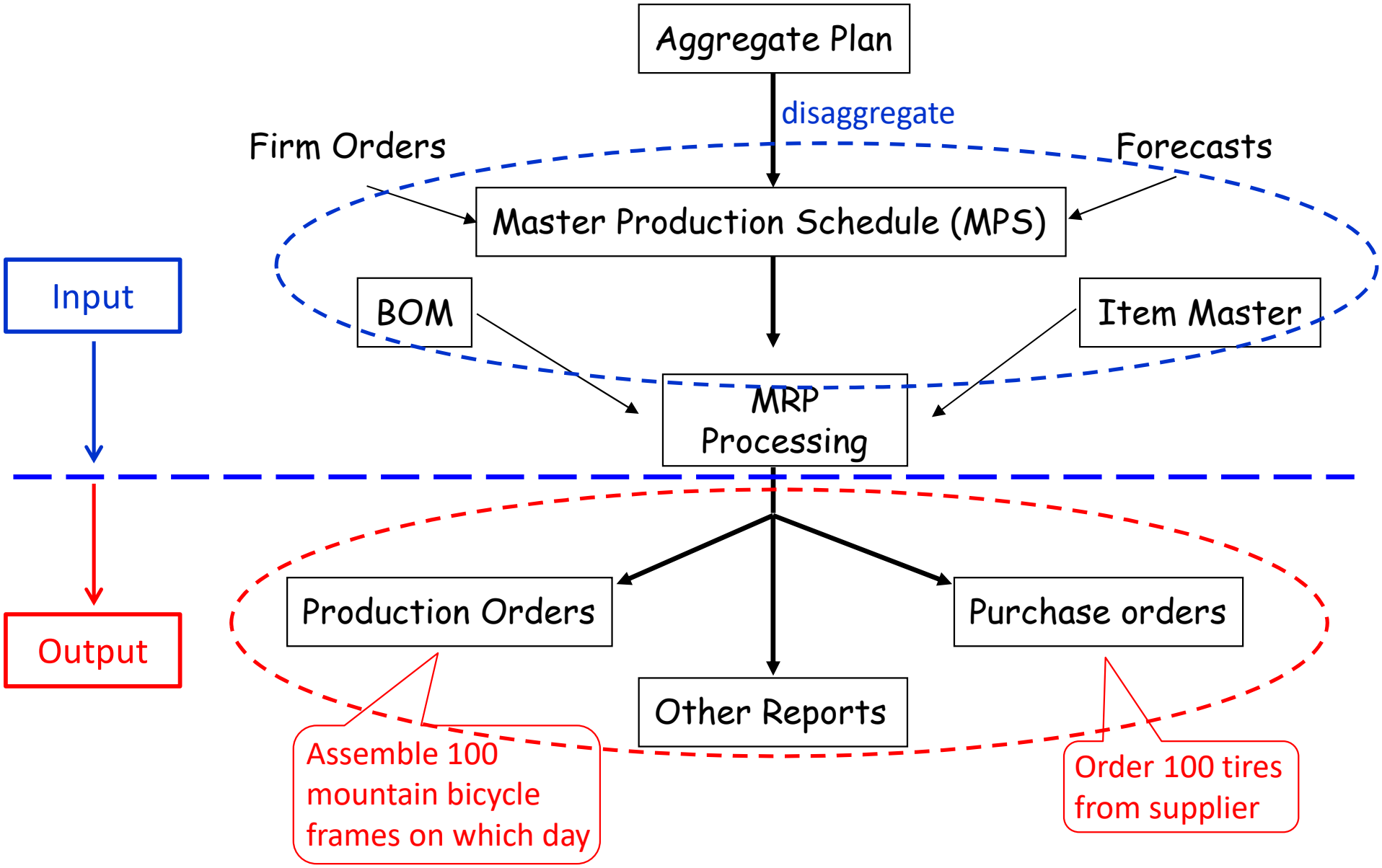


Overview of MRP



Aggregate Plan: Production plan (or staffing plan) made at an aggregate level (or product family), not Stock Keeping Unit (SKU) level (or individual product)

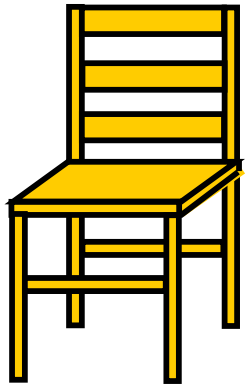
Overview of MRP



MRP Input 1: Master Production Schedule (MPS)

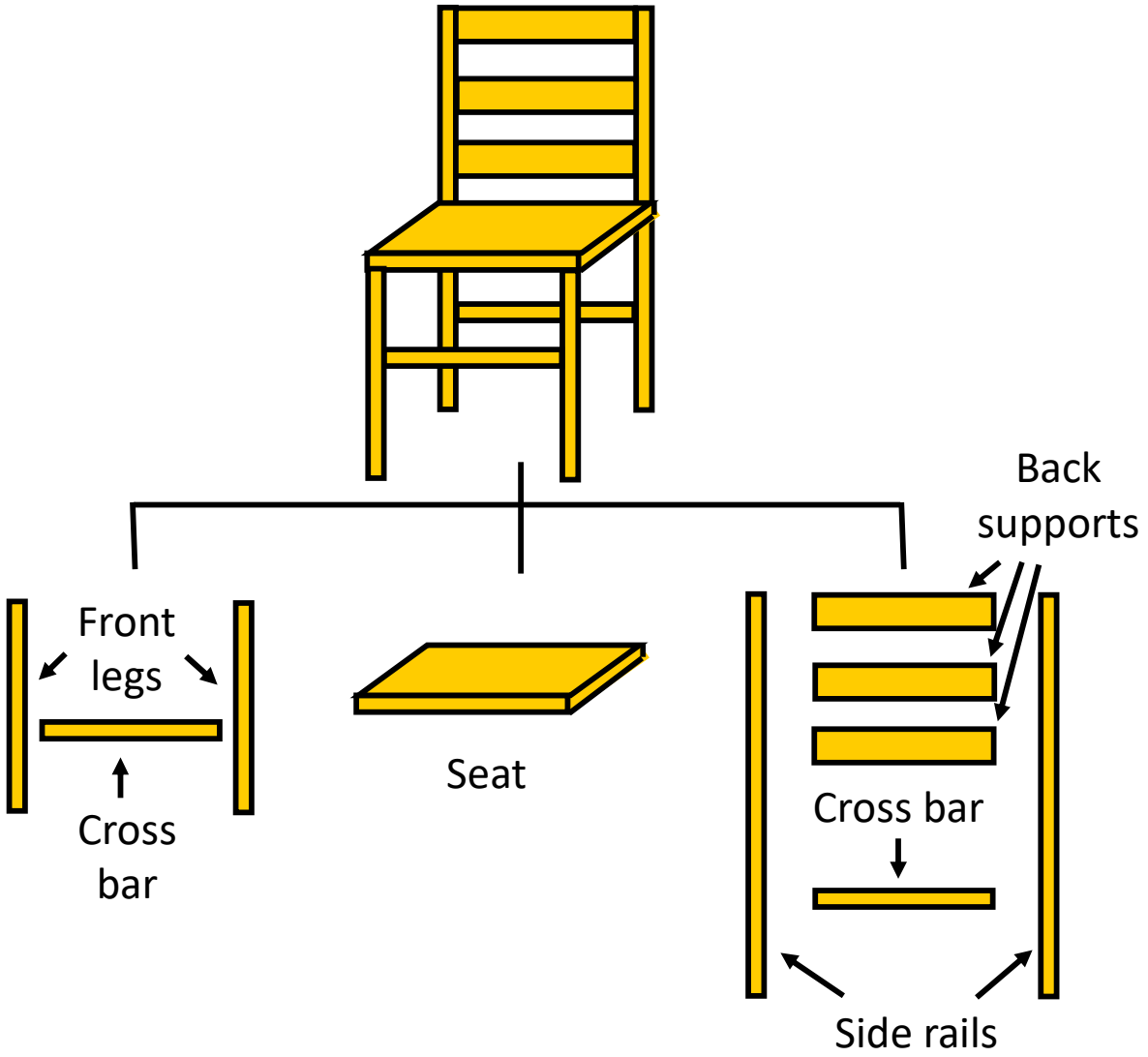
- Master Production Schedule (MPS)
  - States which end items are to be produced, when these are needed, and in what quantities; disaggregated demand from aggregate plan.
  - e.g.

Week	1	2	3	4	5	6	7	8
Chair	600	600	600	600	1000	1000	1000	1000



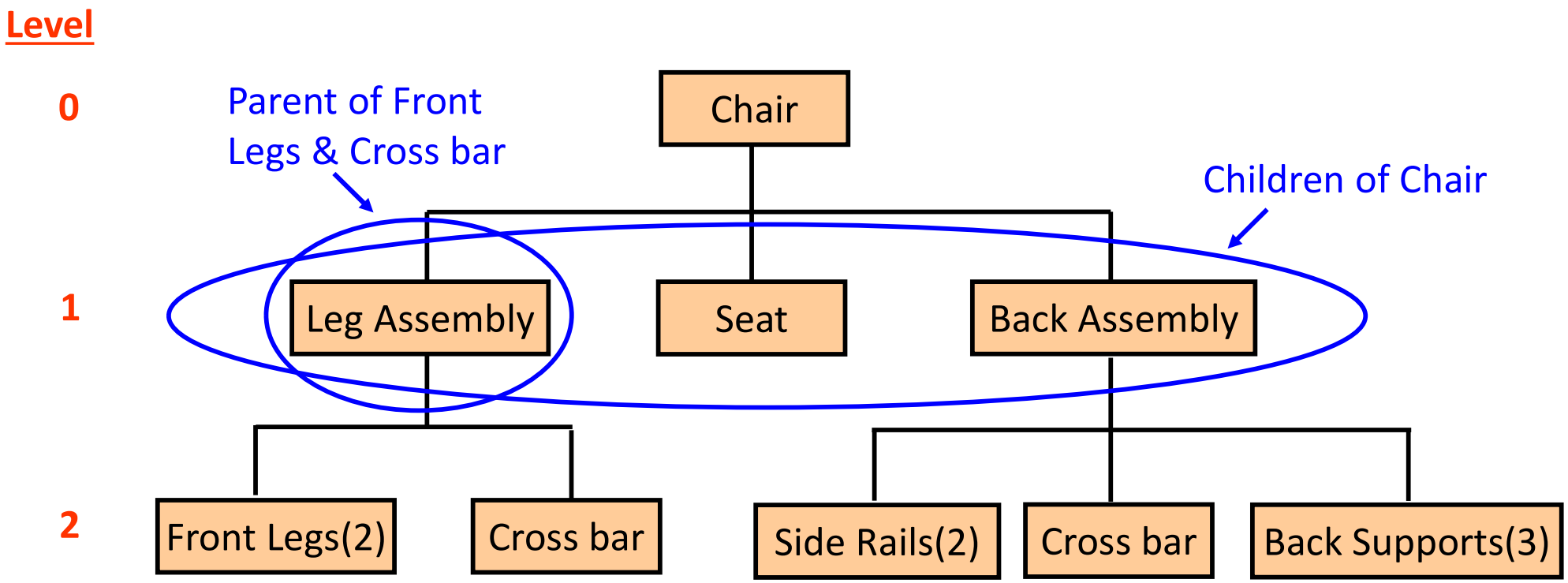
MRP Input 2: Bill of Material (BOM)

- A listing of all the raw materials, parts, subassemblies, and assemblies needed to produce one unit of a product.
- Address What & How Much?
- e.g., Chairs consist of
  - 1 seat;
  - 2 vertical legs;
  - 2 horizontal cross bars;
  - 3 back supports; and,
  - 2 side rails.



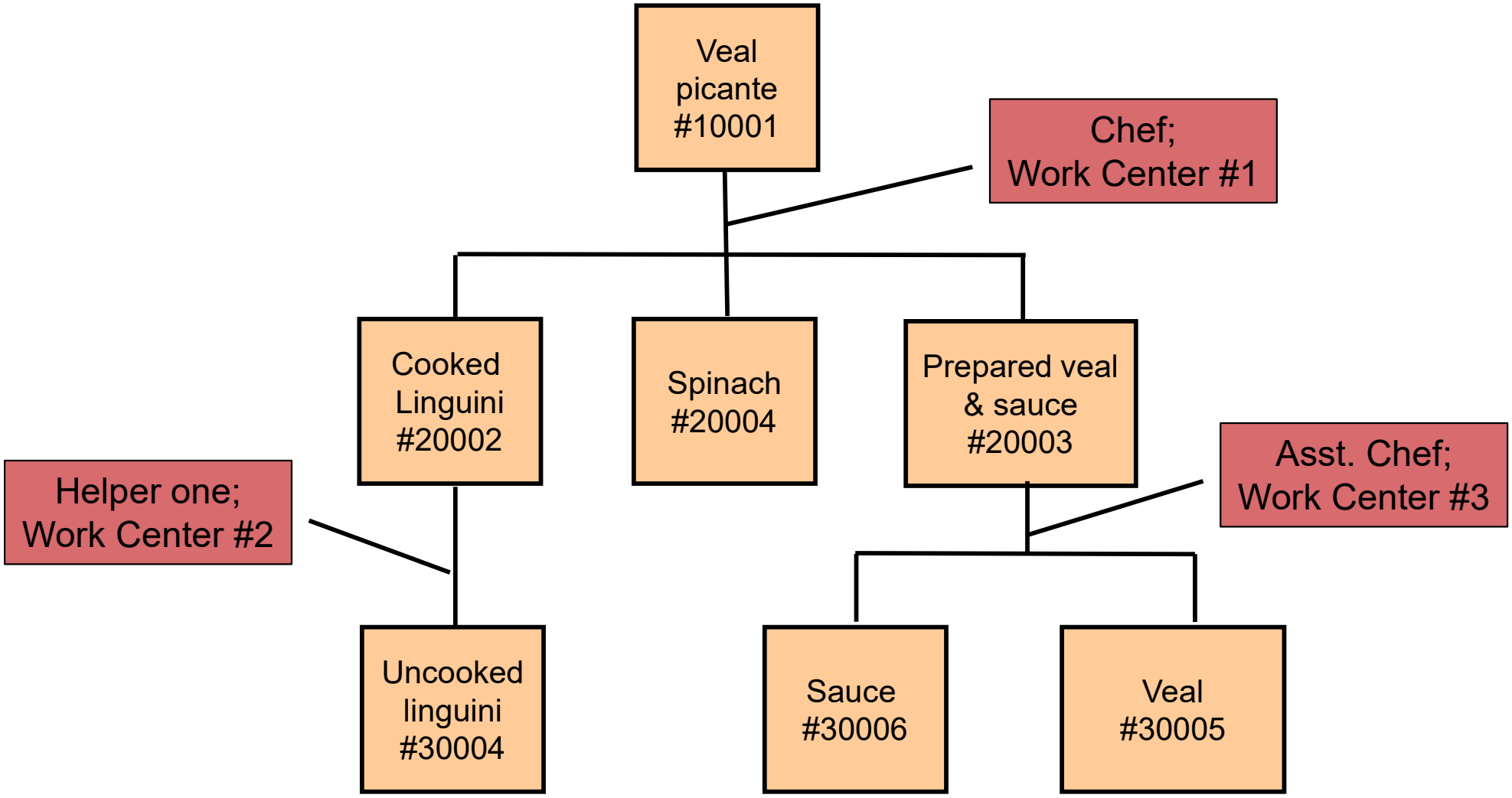
BOM – Product Structure Tree

- *Product Structure Tree* : Visual depiction of the requirements in a bill of materials, where all components are listed by levels.



- Parent-child relationship: When two items are connected to each other in the product structure, an item appearing at the higher level is called the “parent” item and the one at the lower level is the “child”.
- The concept of level: Calculations for the production of items are done in strict sequence of level: from end items to leaves (based on parent-child relationship).

Veal Picante – Product Structure Tree



Source: Adapted from John G. Wacker, "Effective Planning and Cost Control for Restaurants," Production and Inventory Management (Vol. 26, no. 1): 60

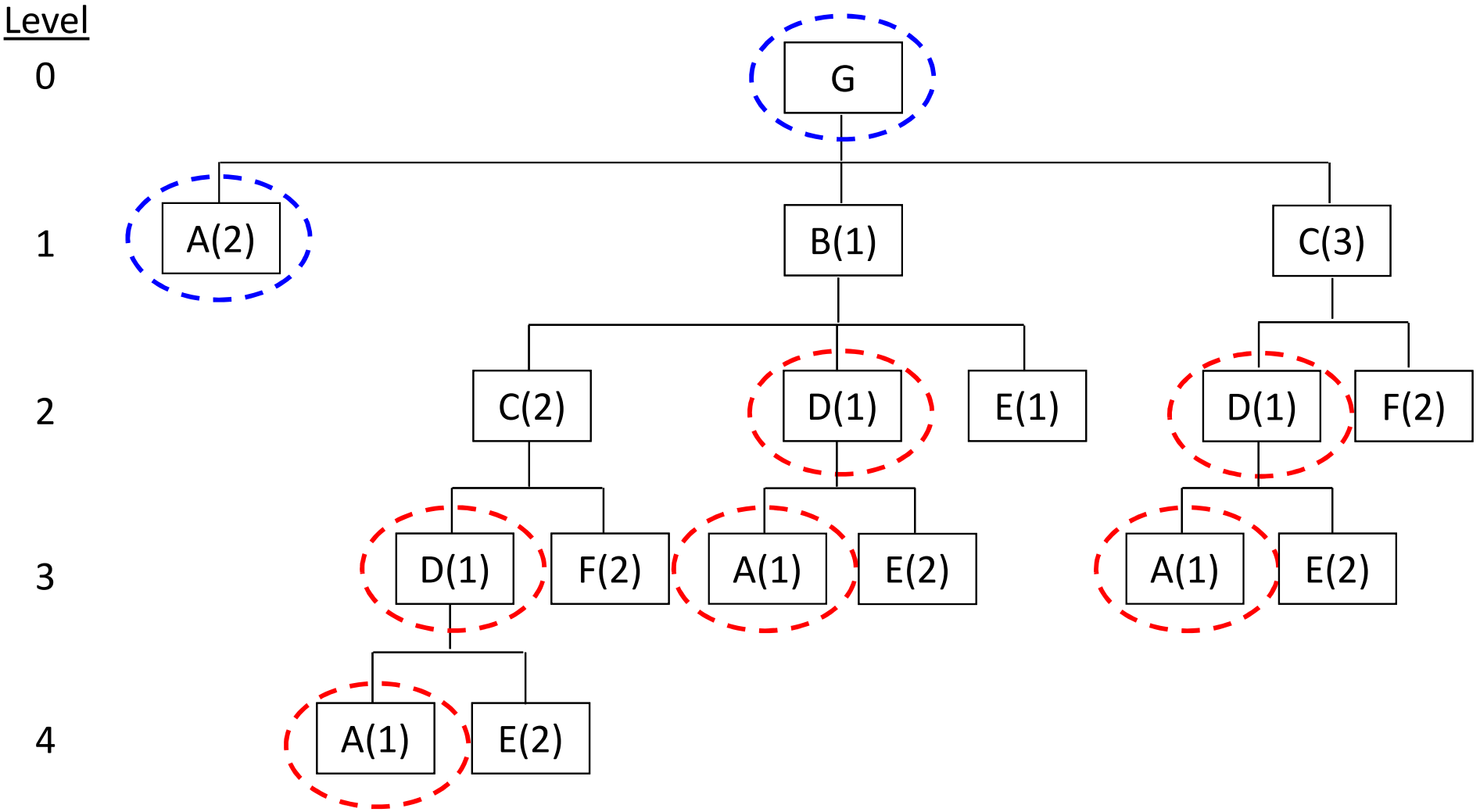
BOM – Tabular Form

- Sometimes, BOM may also be expressed in tabular form.

Level	Item	Description	Child (Multiplicity)
0 - - -	A	Chair	B(1), C(1), D(1)
- 1 - -	B	Leg Assembly	E(2), F(1)
- - 2 -	E	Front Legs	-
- - 2 -	F	Cross bar	-
- 1 - -	C	Seat	-
- 1 - -	D	Back Assembly	G(2), F(1), H(3)
- - 2 -	G	Side rails	-
- - 2 -	F	Cross bar	-
- - 2 -	H	Back support	

- Here, multiplicity refers to the number of units a component (child) is needed for a unit of its parent.
  - e.g., each unit of A requires 1 unit of B; hence the multiplicity of B is 1 for A

The Case of Multiple Parents



- "A" is a child of G and D. What about its multiplicity?
  - The multiplicity of A can be different for different parents (2 for G, 1 for D);
  - but it is always 1 for D, regardless of whether D is for item B or item C.
- Item level is unique – e.g., A belongs to level 4 (lowest level at which it appears).

MRP Input 3: Item Master

- Includes information on the status of each item by time period; e.g. scheduled receipts, on hand inventory, lead times, lot sizing rule, etc.

Item	Description	On-hand Inventory	Lead Time	Lot Sizing
Chair	A	10	2	L4L
Leg Assembly	B	0	2	L4L
Front Legs	E	6	1	FOQ 50
Cross bar	F	10	1	L4L
Seat	C	3	2	FOQ 40
Back Assembly	D	1	2	L4L
Side rails	G	5	2	L4L
Back support	H	3	3	L4L

Lead Time (When?)

- Chairs are needed in 7 days

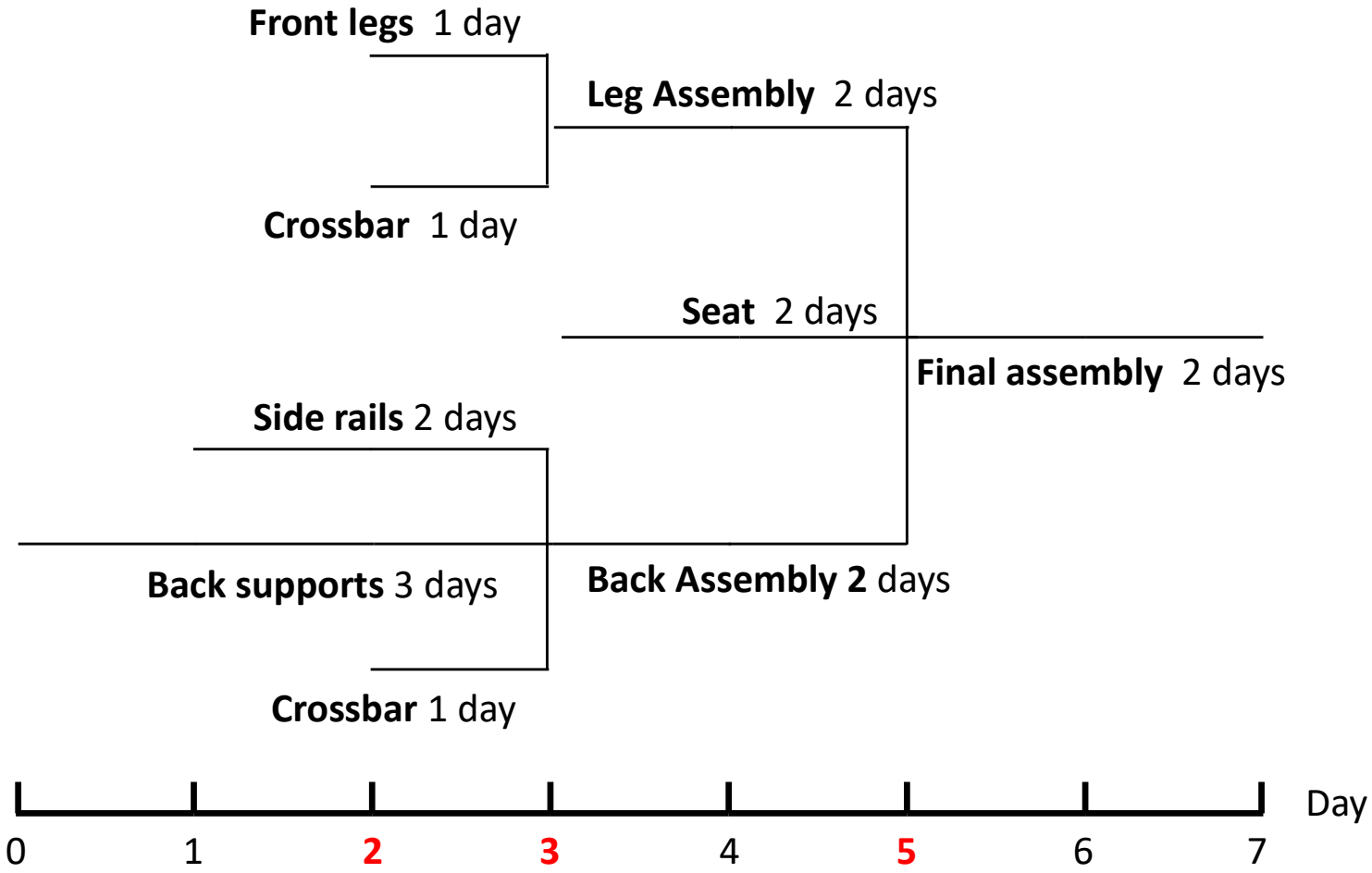


Figure 1: Assembly process

Back Supports: Lead time = 3 Periods; Multiplicity = 3

Assume OH inventory to begin = 270

Period	1	2	3	4	5	6	7	8	9	10
Back Assembly Production Schedule	**90**	**0**	**0**	**0**	**80**	**20**	**0**	**0**	**50**	**70**
Back Support Production Schedule	**0**	**240**	**60**	**0**	**0**	**150**	**210**	**0**	**0**	**0**

Lot Sizing

- Lot-sizing (rules used to determine how to meet requirement)
  - Lot for lot (L4L): Order as needed to meet demand for the period.
  - Fixed Order Quantity (FOQ): Order a fixed quantity. If the FOQ value is 50 and (net) demand is 35, you would still order 50 units. If demand exceeds 50, then order multiples of 50.
    - Why FOQ?

Exercise – Impact of Lot Sizing (Lead Time = 0)

Period		1	2	3	4	5	6	7	8	9	10
Qty. Needed		**20**	**40**	**30**	**0**	**80**	**20**	**0**	**0**	**50**	**70**
L4L: Production		**20**	**40**	**30**	**0**	**80**	**20**	**0**	**0**	**50**	**70**
FOQ 50	Production	**50**	**50**								
	Ending Inv.	**30**	**40**								



40 = 30+50–40

MRP Explosion – Terminology

- Gross Requirements (GR)
  - Total expected demand
- Scheduled Receipts (SR)
  - Incoming orders scheduled to arrive
- On Hand inventory (OH)
  - Expected inventory on hand at the end of each time period
- Net Requirements (NR)
  - Actual amount needed in each time period
- Planned Receipts (PR)
  - Net Requirements adjusted for lot-sizing rule
- Planned Order Release (POR)
  - Planned amount to order or produce in each time period

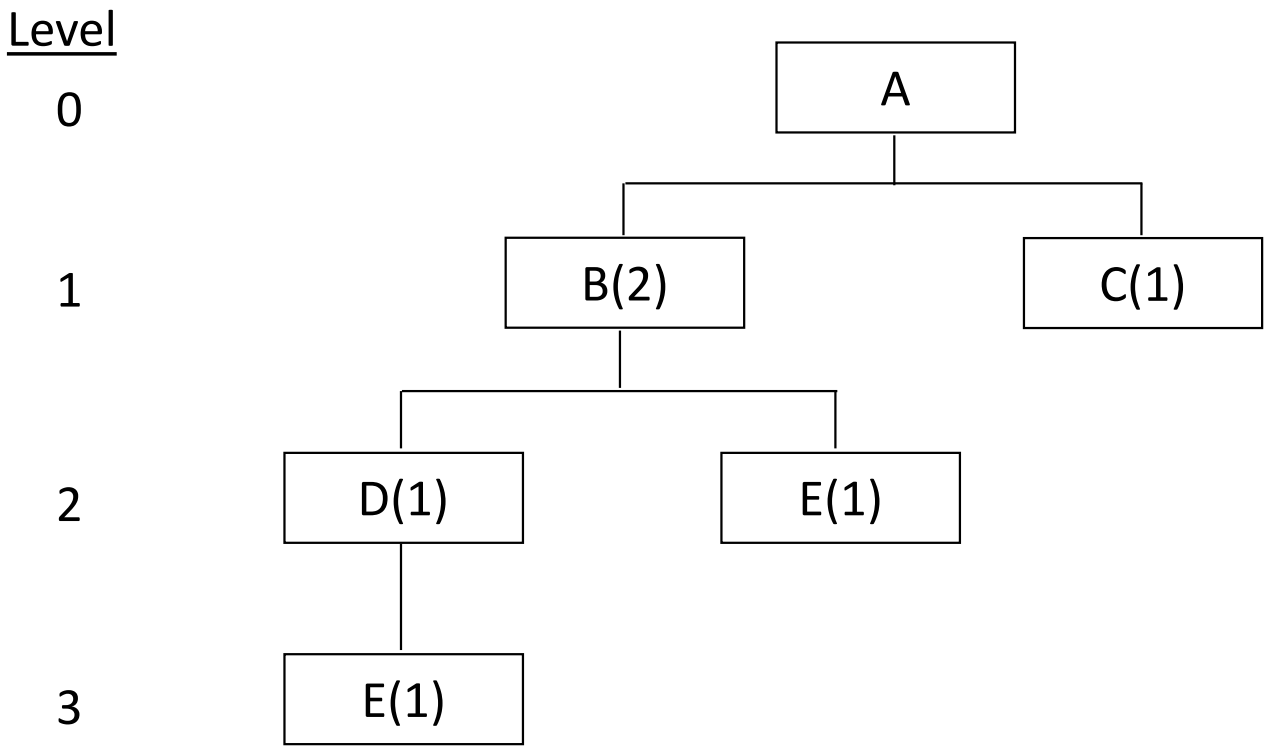
Example 1

Perform MRP calculations for an end item A and its components. The master production schedule (MPS) of an end item A, as well as information from the item master file and the BOM are given as follows:

Week	1	2	3	4	5	6	7	8	9	10
MPS of A	30	60		80	200	120	30		80	70

Item	**Lead Time (wk)**	**Lot Sizing**	**On-hand Inventory**	**Schedule Receipt**	**Child (Multiplicity)**
A	1	L4L	-	SR(1) = 30	B(2), C(1)
B	2	FOQ 250	150	-	D(1), E(1)
C	1	L4L	250	-	-
D	2	FOQ 600	270	SR(2) = 600	E(1)
E	1	L4L	50	SR(1) = 800	-

Example 1: Product Structure Tree



Example 1 : MRP Explosion

Black = Figure given in the question

Blue = Calculated figure

Item		0	1	2	3	4	5	6	7	8	9	10
A L=1 L4L	GR		30	60	0	80	200	120	30	0	80	70
	SR		30									
	OH		0	0								
	NR		0	60	0	80	200	120	30	0	80	70
	PR		0	60	0	80	200	120	30	0	80	70
	POR		60	0	80	200	120	30	0	80	70	
B L=2 FOQ 250 A → 2	GR											
	SR											
	OH	150										
	NR											
	PR											
	POR											
C L=1 L4L A → 1	GR											
	SR											
	OH	250										
	NR											
	PR											
	POR											

NR= 60 to satisfy GR 60

L4L: PR= NR = 60

L=1

Example 1 : MRP Explosion

Black = Figure given in the question

Blue = Calculated figure

Item		0	1	2	3	4	5	6	7	8	9	10
A	GR		30	60	0	80	200	120	30	0	80	70
L=1	SR		30									
L4L	OH		0	0								
	NR		0	60	0	80	200	120	30	0	80	70
	PR		0	60	0	80	200	120	30	0	80	70
	POR		60	0	80	200	120	30	0	80	70	
B	GR		120		160	400	240	60		160	140	
L=2	SR											
FOQ 250	OH	150	30	30	120	220	230	170	170	10	120	120
A → 2	NR				130	280	20				130	
	PR				250	500	250				250	
	POR		250	500	250				250			
C	GR											
L=1	SR											
L4L	OH	250										
A → 1	NR											
	PR											
	POR											

NR= 60 to satisfy GR 60

L4L: PR= NR = 60

L=1

A → B(2)

120= 2*60

30 = 150-120

130 = 160-30

120 = 250 +30-160

FOQ : multiples of 250

NR =130 but FOQ= 250

Example 1 : MRP Explosion

Black = Figure given in the question

Blue = Calculated figure

Item		0	1	2	3	4	5	6	7	8	9	10
A	GR		30	60	0	80	200	120	30	0	80	70
L=1	SR		30									
L4L	OH		0	0								
	NR		0	60	0	80	200	120	30	0	80	70
	PR		0	60	0	80	200	120	30	0	80	70
	POR		60	0	80	200	120	30	0	80	70	
B	GR		120		160	400	240	60		160	140	
L=2	SR											
FOQ 250	OH	150	30	30	120	220	230	170	170	10	120	120
A → 2	NR				130	280	20				130	
	PR				250	500	250				250	
	POR		250	500	250				250			
C	GR		60		80	200	120	30		80	70	
L=1	SR											
L4L	OH	250	190	190	110	0						
A → 1	NR					90	120	30		80	70	
	PR					90	120	30		80	70	
	POR				90	120	30		80	70		

NR= 60 to satisfy GR 60

L4L: PR= NR = 60

L=1

A → B(2)

120= 2*60

30 = 150-120

130 = 160-30

120 = 250 +30-160

FOQ : multiples of 250

NR =130 but FOQ= 250

90 = 200-110

Example 1 : MRP Explosion (cont'd)

Black = Figure given in the question

Blue = Calculated figure

Item		0	1	2	3	4	5	6	7	8	9	10
B	GR		120		160	400	240	60		160	140	
	SR											
L=2	OH	150	30	30	120	220	230	170	170	10	120	120
FOQ 250	NR		0		130	280	20				130	
A → 2	PR		0		250	500	250				250	
	POR		250	500	250				250			
D	GR		250	500	250				250			
	SR			600								
L=2	OH	270	20	120	470	470	470	470	220	220	220	220
FOQ 600	NR				130							
B → 1	PR				600							
	POR		600									
E	GR		850	500	250				250			
	SR	250 +600	800									
L=1	OH	50	0									
L4L	NR			500	250				250			
B → 1	PR			500	250				250			
D → 1	POR		500	250				250				

MRP General Procedure:

Start with period 1 and proceed to the last period. **At period k**,

Step 1 Gross requirement (GR):

For every item that is the parent of the current item, multiply by multiplicity m to the POR value of the parent just calculated in the previous step for period k. Add these values together for all parents of the child item to find the GR value of the child item for period k.

Step 2 Net Requirement (NR):

$$NR = MAX \{GR - SR - OH \text{ for period } (k-1), 0\}$$

Step 3 Calculate Planned Receipt (PR) based on the lot-sizing method:

L4L: PR = NR.

FOQ: PR should be a multiple of FOQ so that the lot size is at least equal to NR. (For example: If NR = 60, and FOQ = 100, PR should be 100. If NR = 225 then PR should be 300 and so on.)

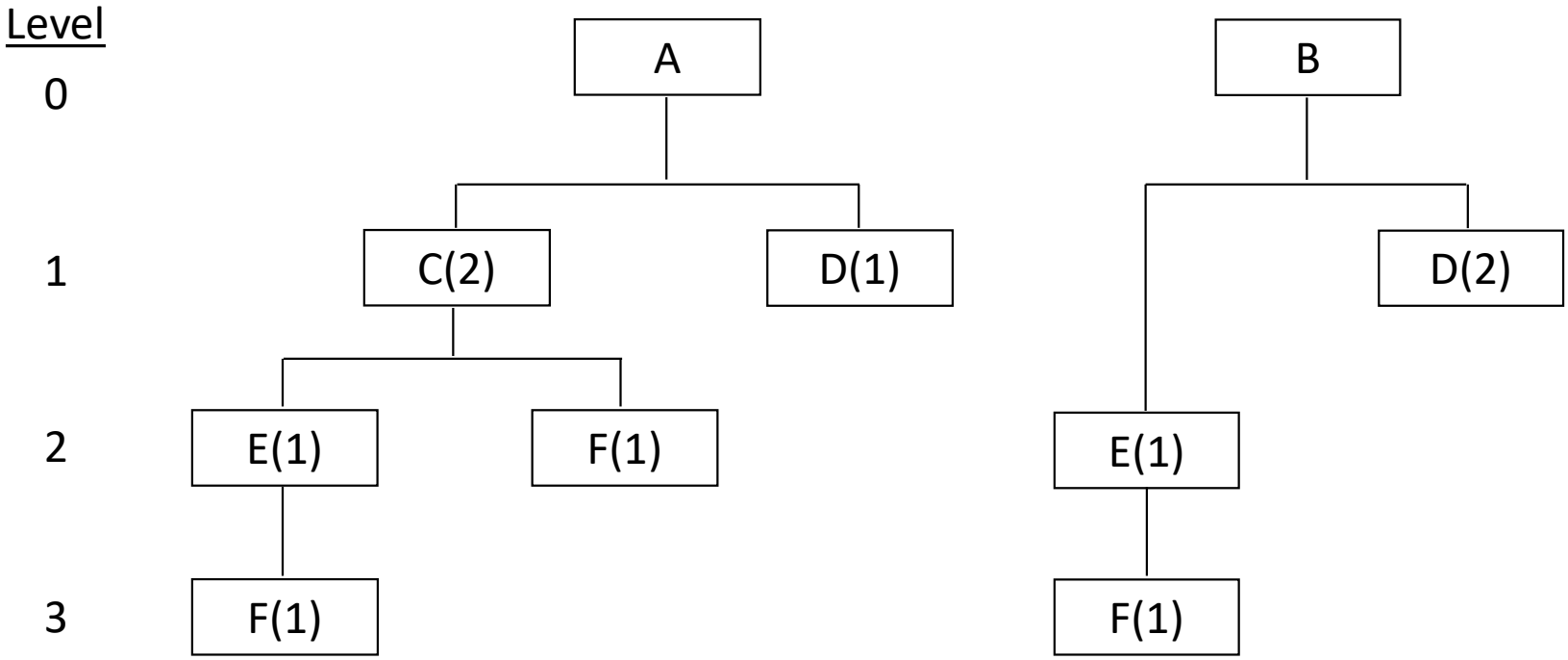
Step 4: On-hand Inventory (OH):

$$OH = OH \text{ for period } (k-1) + SR + PR - GR$$

Step 5: Planned Order Release (POR) = PR for period (k+L), where L is the lead time.

MRP Explosion is Complicated By …

- Safety Stock Requirement (not tested)
  - Why do we need safety stock?
- Other Lot Sizing Rules (not tested)
- Multiple End Items Sharing Common Components
  - e.g., Practice Problem 8, Question 4



MRP Outputs

- Production and/or purchasing schedule – Order releases
  - Schedule indicating the amount and timing of future orders
  - Authorization for the execution of planned orders
- Reschedule notices & exception reports
  - Revisions of due dates or order quantities, expedited orders or cancellations of orders (Why?)
  - Late & overdue orders, reporting errors, excessive scrap rates, and requirement for nonexistent parts
- Performance-control reports
  - Evaluation of system operation
  - Measure deviations from plans (missed deliveries & stockouts); provide information to assess cost performance

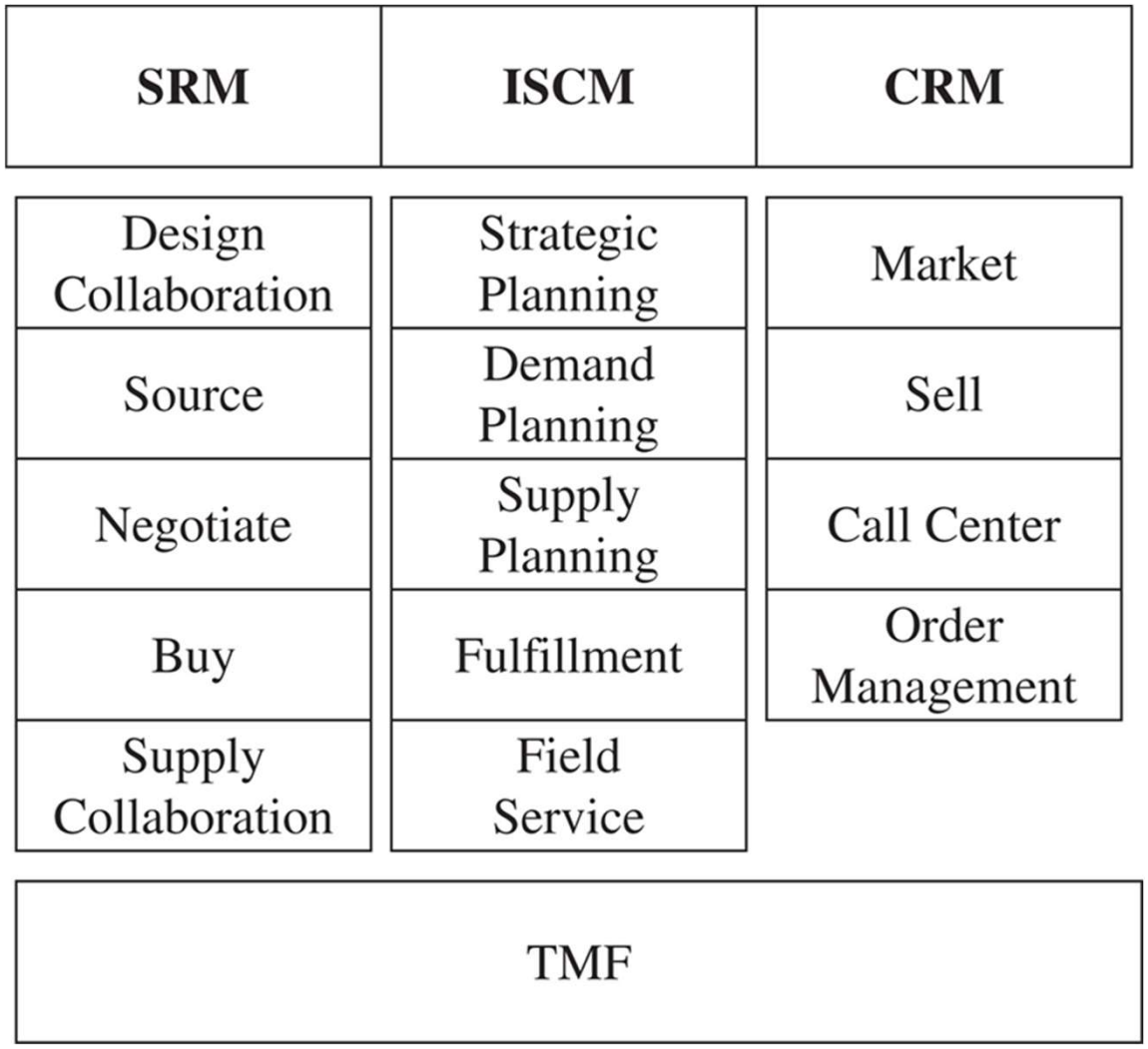
ERP Evolution

- MRP
  - Materials only
- MRP II
  - Includes Capacity Requirements Planning (CRP)
  - Expands to accounting, marketing, etc.
  - Consider accounts payable, accounts receivable, payroll, financial analysis, sales analysis, inventory management, etc.
- ERP
  - Allows more open exchange of data
  - Include cost accounting & management, budgeting, HR activities (training, payroll, retirement & pension plan), project management, Customer Relationship Management (CRM)
  - Consider multiple sites, global customers, languages, currency rates, etc
  - Naively put, ERP = MRP II + CRM (Customer Relationship Management)
  - ERP incorporates the principles of global supply chain management

ERP System Modules

- Three Macro Processes
  - Customer Relationship Management (CRM)
  - Internal Supply Chain Management (ISCM)
  - Supplier Relationship Management (SRM)
- These macro processes rest on a transaction management foundation (TMF), which includes basic ERP and its components such as financials and human resources, database, storage, security, infrastructure, and integration software.
- When enterprise performance is closely linked to supply chain performance, firms must focus on these macro processes instead of just its internal process.

ERP System Modules



Benefits of ERP

- Better performance via efficient & effective usage of organization's resources
  - Lower inventory, higher workforce efficiency and better customer service (providing one source for billing and relationship tracking)
- Effective platform for automation, dynamic control and synchronization of activities (beyond production & inventory planning) & sharing of global & instantaneous information throughout the organization
- Facilitate management & analysis of business processes within an interactive environment

ERP in Practice

- ERP systems can boost your bottom line. But an ineffective ERP implementation can bring a company to its knees.
  - Invacare (medical care product manufacturer) lost US$30 million in 2005 due to problems with Oracle ERP implementation.
  - Vodafone was fined £4,6 million in 2016 by British telecom regulator (Ofcom) for failing to credit customers' accounts after they paid to 'top-up' their mobile phone credit (caused by ERP system failure).



Regulator Ofcom has fined Vodafone £4.6m for "serious" breaches of consumer protection rules, its largest fine to date for a telecoms operator.

~ www.bbc.com, 26 Oct. 2016

- https://www.cio.com/article/2429865/enterprise-resource-planning/enterprise-resource-planning-10-famous-erp-disasters-dustups-and-disappointments.html

ERP Pitfalls

- Assume Integrity of Data
  - Facilitated by mobile technology but systemic problems such as inaccurate or unreliable lead time quotation remain – some ERP systems don't assume this and add Capacity Requirement Planning (CRP) to mitigate this problem
- Poor Implementation Strategy
  - Require Change Management for transition to ERP – e.g. executive involvement, training, total employee involvement, clear goal/expectation
  - Interrupted by departure of sponsor – average tenure of Managing Director is 4 years while average implementations duration is 18 months
- Lack of Adequate Resources
  - Companies fail to consider the true total costs of running ERP – maintenance renewal cost is generally between 15%-22% of the initial software cost
  - Vendors are typically service-led; can't survive only on initial sale value, but also depend on on-going 'value added services', training and consultancy
- Customization
  - Companies must know what they are implementing and what they really need
- Inexperienced Consultants/Over-reliance on Consultants

ERP Trends

- Cloud ERP (Cloud Computing Platform)
  - Delivered using SaaS (Software-as-a-Service) or in general XaaS (Everything-as-a-Service) business model
    - Services offered over the internet
  - Lower upfront costs
    - computing resources leased (subscription-based) versus purchased outright and maintained on premises; ideal for SME
  - Real-time accessibility with backup and disaster recovery plans
  - Ability to adjust amount of cloud service according to need
    - access to state-of-the-art ERP technology
  - Challenges: Loss of control; reliance on provider's security
- Mobile ERP
  - Improved data quality & greater visibility (of activities & demand) via mobile devices
  - System agility & scalability
    - simplifies tiring, long implementations, making process more efficient and more cost effective

ERP Trends (cont'd)

- Two-tier ERP
  - One larger (legacy) system deployed at the corporate level and one smaller system managed at the subsidiary level
  - One-size-fits-all system is economically not viable or operationally inefficient or inflexible (especially for smaller or newly acquired subsidiary)
  - Need to weigh benefits of an integral system versus cost, flexibility & suitability; if possible, stick to one ERP vendor
- Social ERP
  - Integrate social media into ERP system (e.g. incorporate LinkedIn into HR activities, Facebook & Twitter into Marketing activities)
  - Establish in-house social ERP apps mirroring external social media but with interface secured and maintained with own ERP solution.
  - Facilitate collaboration and communication within business eco-system by promoting interaction at more personal level
  - Usefulness?
- Incorporate Big Data Analytics, Internet of Things, Artificial Intelligence etc.